SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario-Based

Code of Conduct:

I P Surendra Kumar Reddy (192425224) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P Surendra Kumar Reddy

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.

3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Question 1

Given

Network Address = **192.168.100.0/24**

Department	**Subnet**
School of Computing	/25
School of Electronics	/26
School of Mechanical Engineering	/27

Step 1: School of Computing (/25)

Subnet Mask: 255.255.255.128

Number of addresses

$$2^{32-25} = 128$$

Usable Hosts

$$128 - 2 = 126$$

Particular	**Address**
Network Address	**192.168.100.0**
First Host	**192.168.100.1**
Last Host	**192.168.100.126**
Broadcast Address	**192.168.100.127**
Usable Hosts	**126**

Explanation: A /25 subnet contains 128 addresses. After reserving the network and broadcast addresses, 126 addresses remain for hosts.

Step 2: School of Electronics (/26)

Subnet Mask: 255.255.255.192

Particular	**Address**
Network Address	**192.168.100.128**
First Host	**192.168.100.129**
Last Host	**192.168.100.190**
Broadcast Address	**192.168.100.191**
Usable Hosts	**62**

Step 3: School of Mechanical Engineering (/27)

Subnet Mask: 255.255.255.224

Particular	**Address**
Network Address	**192.168.100.192**
First Host	**192.168.100.193**
Last Host	**192.168.100.222**
Broadcast Address	**192.168.100.223**
Usable Hosts	**30**

Remaining Unused IP Address Range

192.168.100.224 – 192.168.100.255

Final Answer

Department	Network Address	Host Range	Broadcast	Usable Hosts
Computing	192.168.100.0	192.168.100.1 – 192.168.100.126	192.168.100.127	126
Electronics	192.168.100.128	192.168.100.129 – 192.168.100.190	192.168.100.191	62
Mechanical	192.168.100.192	192.168.100.193 – 192.168.100.222	192.168.100.223	30

Question 2

Given

Network = **10.10.0.0/16**

Department	**Subnet**
Software Development	/24
Quality Assurance	/25
Human Resources	/26
Executive Management	/27

Software Development (/24)

Particular	**Address**
Network Address	**10.10.0.0**
Host Range	**10.10.0.1 – 10.10.0.254**
Broadcast	**10.10.0.255**
Usable Hosts	**254**

Quality Assurance (/25)

Particular	**Address**
Network Address	**10.10.1.0**
Host Range	**10.10.1.1 – 10.10.1.126**
Broadcast	**10.10.1.127**

Usable Hosts	**126**

Human Resources (/26)

Particular	**Address**
Network Address	**10.10.1.128**
Host Range	**10.10.1.129 – 10.10.1.190**
Broadcast	**10.10.1.191**
Usable Hosts	**62**

Executive Management (/27)

Particular	**Address**
Network Address	**10.10.1.192**
Host Range	**10.10.1.193 – 10.10.1.222**
Broadcast	**10.10.1.223**
Usable Hosts	**30**

Remaining Unused Range

10.10.1.224 – 10.10.255.255

Question 3

Given IPv6 Address

2001:0db8:abcd:0000:0000:0000:0000:0000/64

Compressed Form

2001:db8:abcd::/64

Expanded Form

2001:0db8:abcd:0000:0000:0000:0000:0000

Network Prefix

2001:db8:abcd:0::/64

Number of Host Addresses

Host bits

$$128 - 64 = 64$$

Total Hosts

$$2^{64}$$

= 18,446,744,073,709,551,616 host addresses

Explanation

In IPv6, a **/64 prefix** reserves the first 64 bits for the network and the remaining 64 bits for host addresses.

Question 4

Given Routing Table

Network
192.168.1.0/24
192.168.2.0/24
192.168.3.0/24

Destination IP

192.168.2.45

Solution

1. Destination Subnet

192.168.2.0/24

2. Matching Routing Table Entry

192.168.2.0/24

3. Next-Hop Network

Forward the packet through the interface connected to the **192.168.2.0/24** network.

4. Default Route

If no routing table entry matches,

0.0.0.0/0

is used as the **default route**.

Explanation

Routers always select the **longest matching prefix**. If no match exists, the packet is forwarded using the default route.

Question 5

Given

Administration LAN = **192.168.10.0/26**

Finance LAN = **192.168.10.64/26**

Administration Gateway = **192.168.10.1**

Finance Gateway = **192.168.10.65**

Administration LAN

Particular	**Address**
Network Address	**192.168.10.0**
Host Range	**192.168.10.1 – 192.168.10.62**
Broadcast Address	**192.168.10.63**

Sample Computer IPs

- PC1 → **192.168.10.2**
- PC2 → **192.168.10.3**
- PC3 → **192.168.10.4**

Default Gateway

192.168.10.1

Finance LAN

Particular	**Address**
Network Address	**192.168.10.64**
Host Range	**192.168.10.65 – 192.168.10.126**
Broadcast Address	**192.168.10.127**

Sample Computer IPs

- PC1 → **192.168.10.66**
- PC2 → **192.168.10.67**
- PC3 → **192.168.10.68**

Default Gateway

192.168.10.65